

Smart Floor Cleaning Robot

This paper proposed a smart floor cleaning robot that can move automatically, detect obstacles, and clean the floor using a vacuum and mop system. The main objective of the paper is to reduce manual cleaning work by developing a low-cost autonomous cleaning robot. The robot can move on the floor, avoid obstacles, collect dust using a mini vacuum cleaner, and mop the surface using a motor-based cleaning mop.

The main components used in this robot are Arduino Uno, L293D motor shield, four geared motors, ultrasonic sensors, NodeMCU ESP8266, rechargeable battery, buck converter, wheels, switch, connecting wires, mini vacuum cleaner, water bottle, DC motor, and cleaning mop. Arduino Uno is used as the main controller. The L293D motor shield controls the geared motors. The ultrasonic sensor detects obstacles in front of the robot. NodeMCU ESP8266 sends obstacle data to the cloud. The battery powers the robot, and the buck converter controls voltage for the circuit.

The paper includes a proposed system overview, flowchart, and circuit diagram. The system overview shows Arduino Uno connected with the L293D motor driver, motors, ultrasonic sensor, NodeMCU, Wi-Fi, and cloud. The flowchart explains the robot's working process step by step. The circuit diagram shows the motor driver and ultrasonic sensor connection with Arduino.

The robot works in different stages. First, the user turns on the robot. Then the motor driver activates the wheel motors and the robot starts moving forward. At the same time, the vacuum cleaner and mop start working. While moving, the ultrasonic sensor continuously checks for obstacles. If an obstacle is detected within a certain distance, the robot stops and turns left or right. The NodeMCU sends obstacle detection data to ThingSpeak or cloud storage. This allows the user to observe the obstacle data and analyze the robot's movement.

This paper is aligned with our AI-IoT smart farming rover project because it provides a strong idea for the rover movement system. Our farming rover also needs motors, wheels, motor driver, battery, buck converter, and obstacle detection. Instead of using a vacuum and mop, our rover will carry farming sensors such as soil moisture sensor, temperature sensor, humidity sensor, camera, or spraying system. The obstacle avoidance concept is also important for our rover because it must avoid plants, stones, field boundaries, and other objects. Therefore, this paper supports the mechanical movement, motor control, and obstacle detection part of our project.

